

18th Aug

ICS Lec

Variables

Variables are containers for storing data values, such as numbers and characters, in memory.

Rules for naming a variable:

1. Make it relevant.
 2. Letters and digits are allowed. The name should start with a letter, not a number.
 3. “_” is used for compiler/system user.
 4. “C” is a case-sensitive language.
- If you want to print any integer value, we have to use “%d” → format specifier.
 - \t → tab
 - \n → new line

Garbage Value

```
int a, b, sum;  
a = 6;  
b = 7;  
printf("The sum is %d", a + b);  
printf("The sum is %d", sum); // will give a garbage value
```

This will give a garbage value because the variable sum is not defined. Garbage values can be anything and differ from system to system.

scanf()

In C programming, scanf() is a standard library function used to read formatted input from the standard input stream (typically the keyboard). It allows a program to pause and wait for the user to enter data, which is then stored in specified variables.

We cannot use the variable directly; we have to use “&” (ampersand) before the variable, e.g., &a.

OR

Command Line Argument

- &a → ampersand
- scanf needs an address. For the address, we need to use the ampersand.
- Use it only for reading, not printing, because during printing it will display the address of the variable, not the actual value.